

Adaptive Attitude Control for Small Satellites in Low Earth Orbit

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Abstract—Small satellites in Low Earth Orbit must achieve precise pointing under tight actuator and power limits while rejecting environmental disturbances. This paper studies unified attitude stabilization and reaction wheel momentum management for a CubeSat with three reaction wheels and three magnetorquers. The spacecraft is modeled as a rigid body with quaternion kinematics and nonlinear rotational dynamics, including representative LEO disturbances and a time-varying geomagnetic field. A minimal sensor suite is simulated and processed by an error-state EKF estimating attitude and gyro bias; all controllers operate on these estimates. We compare an adaptive discrete-time LQR formulated in the quaternion tangent error space with a nonlinear constrained MPC solved via augmented-Lagrangian iLQR to explicitly handle actuator saturation. High-fidelity simulations on the TRACE CubeSat evaluate nadir pointing, slews, and parametric uncertainty, illustrating trade-offs between tracking performance, control effort, and constraint handling under coordinated wheel-magnetorquer control.

I. INTRODUCTION

The “New Space” era has witnessed a rapid surge in small satellite deployments, driven by reduced launch costs and the accessibility of Low Earth Orbit (LEO) [1].

However, maximizing the utility of these compact platforms requires agility and pointing precision previously reserved for larger spacecraft, placing stringent performance requirements on the Attitude Determination and Control System (ADCS). To meet these operational demands, control strategies must balance high-precision pointing against actuator limitations and limited onboard energy budgets.

Reaction wheels are the most widely adopted actuators for precision pointing due to their ability to provide continuous and fine torque control [2]. Since reaction wheels stabilize the spacecraft by absorbing angular momentum, they are inherently prone to saturation as they accumulate the secular impulse from persistent environmental disturbances, such as gravity gradients and aerodynamic drag. To prevent a loss of control authority, auxiliary actuators capable of generating external torques are required to desaturate the wheels. While magnetic torquers are a popular solution for this task, they are instantaneously underactuated, capable of generating torque only perpendicular to the local time-varying geomagnetic field vector [2]. Beyond hardware constraints, the mathematical representation of the satellite’s attitude introduces significant complexity. Minimal three-parameter representations, such as Euler angles, suffer from kinematic singularities known as gimbal lock, which can cause controller failure during large-angle maneuvers [3]. While unit quaternions offer a globally non-singular four parameter alternative, they complicate the control design by requiring the preservation of a unit norm constraint and handling the double-cover

property of the quaternion, where the states $\mathbf{q}$ and $-\mathbf{q}$ represent the identical physical orientation.

Quaternion based feedback controllers have been applied to the problem of attitude stabilization very successfully over the years [2], [4], [5], [6]. Stability guarantees are provided using Lyapunov approaches [5], [4], [6]. However usually gains are found using heuristics [4] or through manual tuning. Moreover, applying optimal control techniques like the Linear Quadratic Regulator (LQR) to quaternion kinematics is complicated by the unit norm constraint, which renders the simple linearization uncontrollable. To bypass this, designers frequently revert to reduced error representations that fail to respect the intrinsic geometry of the quaternion manifold, treating the state simply as a vector rather than an element of the unit hypersphere [7], [8], [9]. Most commonly the attitude stabilization problem is treated separately from momentum dumping [2], [4]. This ignores possible contributions from the magnetorquers towards attitude stabilization. This cascaded architecture often relies on fixed-gain controllers that do not account for time-varying environmental conditions.

To address these limitations, this work explores unified control frameworks which integrate attitude stabilization with active momentum management, while respecting the quaternions manifold structure and dealing with the time-varying local geomagnetic field. Specifically an adaptive LQR and a Model Predictive Control (MPC) approach. These controllers are then compared against a baseline classical cascaded feedback controller under nominal pointing, slew maneuvers, and parametric uncertainty in a high-fidelity simulation study, utilizing the TUDSat TRACE CubeSat (Fig. 1) as a reference platform. TRACE is a 2U nanosatellite developed by the TU Darmstadt Space Technology group (TUDSat), scheduled for launch in 2026 with a primary mission to demonstrate independent technologies for rapid object identification and tracking in LEO. Its actuation architecture, comprising of three orthogonal reaction wheels and three magnetic torquers, serves as a representative testbed for challenges addressed in this study.

The remainder of this report is organized as follows: Firstly, the environmental and dynamical models of the satellite are developed. Then Section V describes the design and formulation of the controllers. Section VI presents the simulation results. And finally, Section VII discusses the implications of these results and concludes with recommendations for future work.

II. LEO ENVIRONMENT

Unlike deep space missions, a spacecraft in LEO must actively counteract significant aerodynamic drag and gravity gradient torques to maintain precise pointing. Furthermore, the interaction with the Earth’s magnetic field dictates the

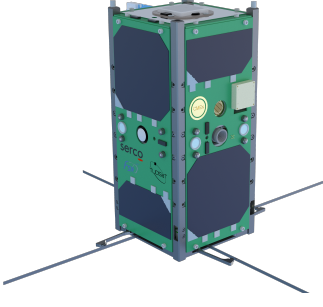


Fig. 1: Reference platform TRACE, a 2U nanosatellite worked on by the TU Darmstadt Space technology group. The satellite is $10 \times 10 \times 22.7$ cm big.

feasibility of magnetorquer actuation, making accurate environmental modeling a prerequisite for effective control design.

A. Gravity Gradient

The gravity gradient torque arises from the variation of Earth's gravitational force across the spacecraft's finite dimensions. Governed by the inverse-square law, the gravitational attraction is stronger on the parts of the satellite closer to the Earth than on those further away. This differential force creates a torque that attempts to align the axis of minimum inertia with the local vertical. For the simulation, this torque τ_{GG} is modeled as

$$\tau_{GG} = \frac{3\mu}{\|\mathbf{r}\|^3} (\hat{\mathbf{r}} \times \mathbf{J} \hat{\mathbf{r}})$$

where μ is Earth's standard gravitational parameter, $\mathbf{r}$ is the geocentric position vector, $\mathbf{J}$ represents the spacecraft's inertia tensor [2]. This acts as a small continuous disturbance that the reaction wheels must absorb to prevent attitude drift.

B. Aerodynamic Drag

Due to the residual atmosphere in LEO, aerodynamic drag constitutes the dominant disturbance force, particularly for CubeSats with large surface-to-mass ratios. To capture the attitude-dependent nature of this disturbance, the spacecraft is modeled as a collection of rectangular surface segments. The total aerodynamic force is the summation of forces acting on each surface i , calculated as

$$\mathbf{F}_{aero,i} = -\rho \|\mathbf{v}_{rel}\|^2 A_i \{ \sigma_{t,i} \hat{\mathbf{v}}_{rel} + (\sigma_{n,i} S_i + (2 - \sigma_{n,i} - \sigma_{t,i}) \cos \theta_i) \hat{\mathbf{n}}_i \} \max(\hat{\mathbf{v}}_{rel}^\top \hat{\mathbf{n}}_i, 0),$$

where ρ is the local atmospheric density derived from the NRLMSISE-00 model [10], $\mathbf{v}_{rel}$ is the velocity vector relative to the atmosphere, and A_i is the surface area. The interaction coefficients $\sigma_{t,i}$ and $\sigma_{n,i}$ account for tangential and normal accommodation, while $\hat{\mathbf{n}}_i$ represents the surface normal unit vector [11]. The resulting disturbance torque is computed by taking the cross product of the force and the vector to the center of pressure $\mathbf{r}_{cop,i}$ for each surface.

C. Solar Radiation Pressure

While generally weaker than aerodynamic drag in low orbits, Solar Radiation Pressure (SRP) introduces a persistent secular disturbance that varies with the sun-synchronous orbit geometry. Similar to the drag model, SRP is computed per surface segment i using

$$\mathbf{F}_{SRP,i} = -PA_i \max(\hat{\mathbf{s}}^\top \hat{\mathbf{n}}_i, 0) \{ (1 - \rho_{s,i} - \rho_{t,i}) \hat{\mathbf{s}} + \left[2\rho_{s,i} \hat{\mathbf{s}}^\top \hat{\mathbf{n}}_i + \frac{2}{3}\rho_{d,i} \right] \hat{\mathbf{n}}_i \},$$

where $P = P(s)$ is the solar radiation pressure constant and $\mathbf{s}$ is the spacecraft to sun vector. The optical properties of the satellite surfaces are defined by the specular $\rho_{s,i}$, diffuse $\rho_{d,i}$, and transmissive $\rho_{t,i}$ reflection coefficients [12]. The simulation strictly accounts for eclipse phases, nullifying this force when the satellite passes through Earth's shadow.

D. Geomagnetic Field

Earth's geomagnetic field, generated by geodynamo processes in the core and magnetization in the crust, varies continuously with the satellite's position and time. This spatial variation is critical for the control architecture, as it dictates the instantaneous control authority of the magnetorquers. To accurately capture these dynamics, the local geomagnetic field vector $\mathbf{b}(t, \mathbf{r})$ is computed using the International Geomagnetic Reference Field (IGRF) model presented in [13].

III. RIGID-BODY DYNAMICS

This section establishes the mathematical framework governing the satellite's translational and rotational motion. We model TRACE as a rigid body orbiting Earth, defining the necessary coordinate frames, orbital mechanics, and attitude dynamics required for accurate simulation.

A. Coordinate Frames

Three primary coordinate frames are defined to describe the satellite's state. The *Earth Centered Inertial* (ECI) frame, denoted $\mathcal{I}$, serves as the global reference with its origin at Earth's center, the Z -axis aligned with Earth's rotation axis, and the X -axis pointing toward the vernal equinox.

For attitude targeting, we define the *Orbit Reference Coordinate* (ORC) frame, also known as Local-Vertical/Local-Horizontal (LVLH) [2], denoted $\mathcal{O}$. Its origin is fixed to the satellite's Center of Mass (CoM). The axes are defined relative to the inertial position $\mathbf{r}_{\mathcal{I}}$ and velocity $\mathbf{v}_{\mathcal{I}}$ such that the Nadir axis $\hat{\mathbf{o}}_{z,\mathcal{I}} = -\hat{\mathbf{r}}_{\mathcal{I}}$ points to the Earth center, the orbit normal $\hat{\mathbf{o}}_{y,\mathcal{I}} = -\hat{\mathbf{r}}_{\mathcal{I}} \times \hat{\mathbf{v}}_{\mathcal{I}}$ opposes the angular momentum vector and $\hat{\mathbf{o}}_{x,\mathcal{I}} = \hat{\mathbf{o}}_{y,\mathcal{I}} \times \hat{\mathbf{o}}_{z,\mathcal{I}}$ completes the right-handed triad and points along the orbit. The transformation of a vector $\mathbf{v}$ represented in the Orbit frame to the Inertial frame is given by $\mathbf{v}_{\mathcal{I}} = \mathbf{R}_{\mathcal{IO}} \mathbf{v}_{\mathcal{O}}$ with $\mathbf{R}_{\mathcal{IO}} = [\hat{\mathbf{o}}_{x,\mathcal{I}}, \hat{\mathbf{o}}_{y,\mathcal{I}}, \hat{\mathbf{o}}_{z,\mathcal{I}}]$

Finally, the *Satellite Body Coordinate* (SBC) frame, $\mathcal{B}$, has its origin in the CoM of the spacecraft like the Orbit frame but is fixed its structure. The control objective is to align $\mathcal{B}$ with $\mathcal{O}$.

B. Orbit Dynamics

The orbital motion is propagated using Newton's law of universal gravitation, augmented with the perturbative accelerations described in Section II. The equation of motion for the satellite's position vector $\mathbf{r}_{\mathcal{I}}$ in the inertial frame is

$$\ddot{\mathbf{r}}_{\mathcal{I}} = -\frac{\mu}{\|\mathbf{r}_{\mathcal{I}}\|^3}\mathbf{r}_{\mathcal{I}} + \mathbf{a}_{\text{pert},\mathcal{I}},$$

where $\mathbf{a}_{\text{pert},\mathcal{I}}$ includes accelerations from J_2 to J_4 zonal harmonics, third-body gravitational forces, aerodynamic drag and solar radiation pressure forces [2].

C. Attitude Dynamics

1) *Kinematics*: The satellite's orientation is parameterized by a unit quaternion $\mathbf{q} \in \mathbb{H}$. Geometrically, unit quaternions reside on a 4-dimensional unit hypersphere. Rotations are composed via quaternion multiplication, and the time-evolution of the attitude is governed by the kinematic differential equation [14]

$$\dot{\mathbf{q}}_{\mathcal{B}} = \frac{1}{2}\Xi(\mathbf{q}_{\mathcal{B}})\boldsymbol{\omega}_{\mathcal{B}}^{\mathcal{BI}}, \quad (1)$$

where $\boldsymbol{\omega}$ is the angular velocity vector of the body frame relative to the inertial frame. The matrix

$$\Xi(\mathbf{q}) := \begin{bmatrix} q_4\mathbf{I}_3 + [\mathbf{q}_{1:3}\times] \\ -\mathbf{q}_{1:3}^\top \end{bmatrix}. \quad (2)$$

is known as the attitude Jacobian. $\frac{1}{2}\Xi(\mathbf{q})$ acts as a mapping from the space of angular velocities ($\mathbb{R}^3$) to the tangent plane of the quaternion hypersphere. This concept of mapping between the curved manifold and its Euclidean tangent space is crucial for the control derivation and is detailed further in Appendix VII-A.

2) *Kinetics*: The rotational kinetics are derived from Euler's rotational equation for a system of rigid bodies. The total angular momentum $\mathbf{h}_{\text{tot}}$ comprises the momentum of the main spacecraft body and the angular momenta of the spinning reaction wheels [15]

$$\mathbf{h}_{\text{tot}} = \mathbf{J}\boldsymbol{\omega} + \sum_{k=1}^N J_{w,k} (\Omega_k + \hat{\mathbf{a}}_k^\top \boldsymbol{\omega}) \hat{\mathbf{a}}_k, \quad (3)$$

where $J_{w,k}$ and Ω_k are the axial inertia and speed of the k -th reaction wheel, and $\hat{\mathbf{a}}_k$ is its spin axis. Here, Ω_k denotes the wheel spin rate relative to the spacecraft body (tachometer output). The rotational equation of motion relates the change in momentum represented in $\mathcal{B}$ wrt. $\mathcal{I}$ to the external torques $\boldsymbol{\tau}_{\text{ext},\mathcal{B}}$ (gravity gradient, magnetic, etc.) and internal control torques:

$$\dot{\mathbf{h}}_{\text{tot},\mathcal{B}} + \boldsymbol{\omega}_{\mathcal{B}}^{\mathcal{BI}} \times \mathbf{h}_{\text{tot},\mathcal{B}} = \boldsymbol{\tau}_{\text{ext},\mathcal{B}}. \quad (4)$$

The inertia tensor $\mathbf{J}_{\mathcal{B}}$ is defined for the locked-rotor spacecraft (bus + wheels treated as rigidly attached), while $\hat{\mathbf{J}}_{\mathcal{B}}$ is the effective bus inertia used after eliminating wheel-axis coupling terms for integration. For the simulation, we introduce a modified inertia matrix

$$\hat{\mathbf{J}}_{\mathcal{B}} = \mathbf{J}_{\mathcal{B}} - \sum_{k=1}^N J_{w,k} \hat{\mathbf{a}}_k \hat{\mathbf{a}}_{k,\mathcal{B}}^\top$$

to decouple the body acceleration from the wheel accelerations, allowing for explicit numerical integration of $\dot{\boldsymbol{\omega}}_{\mathcal{B}}^{\mathcal{BI}}$ by substituting (3) into (4) yielding

$$\dot{\boldsymbol{\omega}}_{\mathcal{B}}^{\mathcal{BI}} = \hat{\mathbf{J}}_{\mathcal{B}}^{-1} (\boldsymbol{\tau}_{\text{ext},\mathcal{B}} - \boldsymbol{\tau}_{w,\mathcal{B}} - \boldsymbol{\omega}_{\mathcal{B}}^{\mathcal{BI}} \times \mathbf{h}_{\text{tot},\mathcal{B}}) \quad (5)$$

where $\boldsymbol{\tau}_{w,\mathcal{B}}$ is the torque produced by the RWs.

IV. SATELLITE

This section summarizes the onboard actuation, sensing, and state-estimation models used in the simulation; unless stated otherwise, vectors are expressed in the body frame $\mathcal{B}$.

A. Actuators

The satellite is actuated by a set of reaction wheels (RW) for precise attitude control and three orthogonal magnetorquers (MTQ) for momentum management. To realistically capture actuator dynamics, both systems are modeled as current-controlled first-order systems.

a) *Reaction Wheels*: The satellite is equipped with 3 reaction wheels. Each wheel i has an axial inertia $J_{w,i}$ and spins about an axis $\hat{\mathbf{a}}_i$ expressed in the body frame. The electrical dynamics of the motor current i_k are modeled as:

$$\frac{di_k}{dt} = \frac{i_{\text{cmd},k} - i_k}{T_{\text{cur},k}}, \quad |i_k| \leq i_{\text{max},k} \quad (6)$$

where $i_{\text{cmd},k}$ is the commanded motor current, $T_{\text{cur},k}$ is the current-loop time constant, and $i_{\text{max},k}$ is the saturation limit. The motor torque $\tau_{w,k}$ is proportional to the current via the torque constant $K_{w,k}$:

$$\tau_{w,k} = K_{w,k} i_k \quad (7)$$

The rotational dynamics of the k -th wheel are governed by the balance of motor torque and the counter-torque induced by the spacecraft's angular acceleration:

$$\dot{\Omega}_k = \frac{\tau_{w,k}}{J_{w,k}} - \hat{\mathbf{a}}_k^\top \dot{\boldsymbol{\omega}}_{\mathcal{B}}^{\mathcal{BI}} \quad (8)$$

where $\boldsymbol{\omega}_{\mathcal{B}}^{\mathcal{BI}}$ is the spacecraft angular velocity of the body frame $\mathcal{B}$ relative to the inertial frame $\mathcal{I}$, expressed in $\mathcal{B}$. Here, Ω_k denotes the wheel spin rate relative to the spacecraft body (tachometer output). The rotor absolute spin rate about $\hat{\mathbf{a}}_k$ is $\Omega_k + \hat{\mathbf{a}}_k^\top \boldsymbol{\omega}$.

b) *Magnetorquers*: produce a magnetic dipole moment $\mathbf{m}$, with dipole magnitude $m_k = K_{m,k} i_{m,k}$ proportional to the coil current $i_{m,k}$. The MTQ coil currents are modeled as current-controlled first-order systems:

$$\frac{di_{m,k}}{dt} = \frac{i_{\text{cmd},m,k} - i_{m,k}}{T_{\text{cur},m,k}}, \quad |i_{m,k}| \leq i_{\text{max},m,k}. \quad (9)$$

The resulting control torque is generated by the interaction with the local geomagnetic field $\mathbf{b}_{\mathcal{B}}$:

$$\boldsymbol{\tau}_{m,\mathcal{B}} = \sum_k (m_k \hat{\mathbf{n}}_k) \times \mathbf{b}_{\mathcal{B}} \quad (10)$$

This torque is constrained to be perpendicular to the magnetic field vector, resulting in instantaneous underactuation.

B. Sensors

The controllers require attitude, angular velocity, and (for momentum management) reaction wheel momentum. In practice these states are not directly available; only noisy, sampled sensor outputs are. Accordingly, the simulation provides a minimal onboard sensor suite whose measurements are the sole inputs to the estimator and controller.

a) Gyroscope.: The gyro measures the body angular rate with additive bias and noise,

$$\boldsymbol{\omega}_{\text{mea}} = \boldsymbol{\omega} + \mathbf{b}_\omega + \mathbf{n}_\omega, \quad \dot{\mathbf{b}}_\omega = \mathbf{n}_b, \quad (11)$$

where $\mathbf{b}_\omega$ is modeled as a random walk. The EKF uses the bias-compensated rate $\tilde{\boldsymbol{\omega}} = \boldsymbol{\omega}_{\text{mea}} - \mathbf{b}_\omega$ to propagate the quaternion between vector updates.

b) Magnetometer and Sun sensor.: The magnetometer and Sun sensor provide body-frame *direction* measurements. We model

$$\mathbf{z}_{\text{mea}} = \mathbf{z}_B + \mathbf{n}_z, \quad \mathbf{z} \in \{\mathbf{b}_B, \mathbf{s}_B\}, \quad (12)$$

and normalize both measurement and prediction before forming the EKF residual. During eclipse, the Sun sensor is unavailable and the corresponding update is skipped.

c) Reaction wheel speed sensing.: Wheel speeds are measured with noise, $\Omega_{\text{mea},i} = \Omega_i + n_{\Omega,i}$, and are used to reconstruct the wheel momentum needed by the controllers,

$$\tilde{\mathbf{h}}_w \approx \sum_{i=1}^3 J_{w,i} (\Omega_{\text{mea},i} + \hat{\mathbf{a}}_i^\top \tilde{\boldsymbol{\omega}}) \hat{\mathbf{a}}_i. \quad (13)$$

C. State Estimation (EKF)

All controllers operate on an estimated state rather than the simulated ground truth. We use an Extended Kalman Filter (EKF) to estimate the attitude quaternion $\tilde{\mathbf{q}}$ and gyroscope bias $\tilde{\mathbf{b}}_\omega$. The prediction step propagates $\tilde{\mathbf{q}}$ using the bias-compensated gyroscope rate $\tilde{\boldsymbol{\omega}}$, while measurement updates use vector observations from the magnetometer and Sun sensor. During eclipse the Sun sensor measurement is unavailable and the corresponding update is skipped. The controller angular-rate input is taken as the bias-compensated gyro measurement $\tilde{\boldsymbol{\omega}} = \boldsymbol{\omega}_{\text{mea}} - \mathbf{b}_\omega$, and wheel momentum is reconstructed from wheel-speed measurements as in Section IV-B. The EKF is applied consistently across all experiments to ensure a fair controller comparison under realistic sensing. Further EKF formulation details and additional estimation plots are provided in the Appendix VII-B.

V. CONTROLLERS

Control design for LEO satellites faces challenges including magnetic underactuation, strict pointing requirements, and actuator saturation. This section presents three different control strategies: first, a robust baseline classical controller is defined; second, an Adaptive LQR is introduced to minimize actuation effort under varying magnetic fields and finally, a non-linear MPC is formulated to extend the adaptive LQR to explicitly handle system constraints and non-linearities.

A. Problem Statement

The control objective is to align the body frame with the orbit frame, ensuring the satellite's z-axis maintains a Nadir-pointing orientation. This requires a commanded relative orientation of the identity quaternion $\mathbf{q}_{BO,c} = [0, 0, 0, 1]^\top$ and a null relative angular velocity $\boldsymbol{\omega}_{B_c}^{BO} = \mathbf{0}$. For a circular orbit, the reference angular velocity

$$\boldsymbol{\omega}_c = \boldsymbol{\omega}_O^{OI} = [0, -\omega_0, 0]^\top$$

aligns with the negative orbit normal, with $\omega_0 = \frac{\|\mathbf{v}\|}{\|\mathbf{r}\|}$ being the orbital mean motion. The time-varying reference quaternion $\mathbf{q}_c(t)$ is obtained by integrating the kinematic equation (1) for $\boldsymbol{\omega}_c$. Furthermore, we design the reference trajectory to be a zero-control equilibrium is sufficient to maintain the Nadir orientation. By substituting the reference angular velocity $\boldsymbol{\omega}_c$ into the dynamics (5) and assuming $\boldsymbol{\omega}_c \approx \mathbf{0}$, we observe that the internal gyroscopic torque vanishes only if the total angular momentum of the system is zero. This means the reference wheel momentum has to be $\mathbf{h}_{w,c} = -\hat{\mathbf{J}}_B \boldsymbol{\omega}_c$.

B. Actuator Mapping and Model Reformulation

The attitude controllers generate torque commands which are then transformed to current commands for the actuators. The electrical dynamics are not considered in the controller design, as they are significantly faster than the rotational dynamics of the CubeSat. We define the magnetorquer torque command $\mathbf{u}_m$ such that the generated torque is

$$\boldsymbol{\tau}_{m,B} = (\mathbf{I}_3 - \hat{\mathbf{b}}_B \hat{\mathbf{b}}_B^\top) \mathbf{u}_m, \quad (14)$$

where $\hat{\mathbf{b}}_B = \frac{\mathbf{b}_B}{\|\mathbf{b}_B\|}$ is the normalized local magnetic field vector. The commanded magnetic dipole is

$$\mathbf{m}_B = \hat{\mathbf{b}}_B \times \frac{\mathbf{u}_m}{\|\mathbf{b}_B\|} = \sum_i m_i \hat{\mathbf{n}}_{i,B}. \quad (15)$$

This mapping ensures that only dipole moments perpendicular to the magnetic field are generated. Parallel components are discarded as they cannot produce torque and would result in wasted energy [8].

The reaction wheel torque command is defined with a reversed sign, $\boldsymbol{\tau}_{w,B} = -\mathbf{u}_w$, such that it behaves as an equivalent external torque on the spacecraft body.

Excluding environmental disturbances, the rotational equations of motion (5) are reformulated for control design as

$$\hat{\mathbf{J}}_B \dot{\boldsymbol{\omega}}_B^{BI} = (\mathbf{I}_3 - \hat{\mathbf{b}}_B \hat{\mathbf{b}}_B^\top) \mathbf{u}_m + \mathbf{u}_w \quad (16)$$

$$- \boldsymbol{\omega}_B^{BI} \times (\hat{\mathbf{J}}_B \boldsymbol{\omega}_B^{BI} + \mathbf{h}_w), \quad (17)$$

where $\mathbf{h}_w$ denotes the total angular momentum of the wheels in the spin direction. Its dynamics are given by

$$\dot{\mathbf{h}}_w = \sum_{i=1}^N J_{w,i} (\dot{\Omega}_i + \hat{\mathbf{a}}_i^\top \dot{\boldsymbol{\omega}}_B^{BI}) \hat{\mathbf{a}}_i = \boldsymbol{\tau}_w = -\mathbf{u}_w. \quad (18)$$

In the following, we adopt the simplified notations $\hat{\mathbf{J}} := \hat{\mathbf{J}}_B$, $\boldsymbol{\omega} := \boldsymbol{\omega}_B^{BI}$, and $\mathbf{q} := \mathbf{q}_{BI}$. The full state of the system

is defined as $\mathbf{x} = [\mathbf{q}^\top, \boldsymbol{\omega}^\top, \mathbf{h}_w^\top]^\top$. Combining the dynamics equations (1), (16) and (18) we get

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{f}_{\text{cont}}(\mathbf{x}, \mathbf{u}) \\ &= \begin{bmatrix} \frac{1}{2}\Xi(\mathbf{q})\boldsymbol{\omega} \\ \hat{\mathbf{J}}^{-1} \left((\mathbf{I}_3 - \hat{\mathbf{b}}\hat{\mathbf{b}}^\top)\mathbf{u}_m + \mathbf{u}_w - \boldsymbol{\omega} \times (\hat{\mathbf{J}}\boldsymbol{\omega} + \mathbf{h}_w) \right) \\ -\mathbf{u}_w \end{bmatrix}. \end{aligned}$$

To get the discrete dynamics $\mathbf{f}_{\text{disc}}(\mathbf{x}, \mathbf{u})$ a Runge-Kutta-4 scheme is applied, and then the quaternion is normalized. This technically is not correct because it doesn't respect the quaternions manifold structure but it is common in practice and yields good results [3].

C. Baseline

The classical approach [2], [4] is cascaded a controller with the inner fast loop using the reaction wheels for attitude stabilization and an outer slower loop using magnetorquers for dumping the angular momentum which naturally builds up in the reaction wheels because of non-conservative disturbance torques. For the reaction wheels, we employ a PD-type control law

$$\mathbf{u}_w = -k_p \text{sgn}(\delta q_4) \delta \mathbf{q}_{1:3} - k_d \delta \boldsymbol{\omega}$$

with the error quaternion $\delta \mathbf{q} = \mathbf{q} \otimes \mathbf{q}_c^{-1}$ and the error angular velocity $\delta \boldsymbol{\omega} = \boldsymbol{\omega} - \boldsymbol{\omega}_c$. The inclusion of the term $\text{sgn}(\delta q_4)$ enforces a shortest-path rotation. If omitted when $\delta q_4 < 0$, the satellite would rotate the "long way" ($> 180^\circ$) to the target, a phenomenon known as unwinding [2]. Simultaneously, the momentum dumping law commands a magnetic dipole moment proportional to the accumulated wheel momentum:

$$\mathbf{u}_m = -k_m(\mathbf{h}_w - \mathbf{h}_{w,c}) = -k_m \delta \mathbf{h}_w.$$

This classical strategy can be interpreted as a linear state feedback controller acting on the reduced error state $\delta \mathbf{x}$:

$$\begin{bmatrix} \mathbf{u}_m \\ \mathbf{u}_w \end{bmatrix} = - \underbrace{\begin{bmatrix} k_q \mathbf{I}_3 & k_\omega \mathbf{I}_3 & k_m \mathbf{I}_3 \end{bmatrix}}_{\mathbf{K}} \underbrace{\begin{bmatrix} \text{sgn}(\delta q_4) \delta \mathbf{q}_{1:3} \\ \delta \boldsymbol{\omega} \\ \delta \mathbf{h}_w \end{bmatrix}}_{\delta \mathbf{x}}.$$

D. Adaptive LQR

While the baseline controller guarantees global stability, its sparse gain structure leaves potential performance improvements unexploited. To utilize the full degrees of freedom of the feedback system, we employ a discrete infinite-horizon LQR. The objective is to minimize the quadratic cost function

$$J = \sum_{k=0}^{\infty} (\delta \mathbf{x}_k^\top \mathbf{Q} \delta \mathbf{x}_k + \mathbf{u}_k^\top \mathbf{R} \mathbf{u}_k)$$

subject to the linearized discrete dynamics $\delta \mathbf{x}_{k+1} = \mathbf{A} \delta \mathbf{x}_k + \mathbf{B} \mathbf{u}_k$, where $\mathbf{Q} \succeq 0$ and $\mathbf{R} \succ 0$ are the state and control weighting matrices respectively. The optimal control law is given by the state feedback

$$\mathbf{u}_k = -(\mathbf{R} + \mathbf{B}^\top \mathbf{P} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{P} \mathbf{A} \delta \mathbf{x}_k = -\mathbf{K} \delta \mathbf{x}_k,$$

where $\mathbf{P}$ is the unique positive-definite solution to the Discrete Algebraic Riccati Equation (DARE).

A direct application of this linear framework to the quaternion state is problematic due to the unit norm constraint, which renders the full 4D linearization uncontrollable. To resolve this, we leverage the geometric structure of the quaternion hypersphere, knowing that the derivatives are vectors $\phi \in \mathbb{R}^3$. We choose the reduced quaternion representation to match the baseline controller, i.e. $\phi = \text{sgn}(\delta q_4) \delta \mathbf{q}_{1:3}$. As discussed in the context of the kinematics equation (1), $\Xi(\mathbf{q})$ acts as the linearization of the mapping between the flat tangent space and the curved quaternion manifold (see Appendix VII-A for the full derivation). The resulting linearized system matrices are

$$\mathbf{A}_k = \mathbf{E}(\mathbf{q}_c)^\top \left. \frac{\partial \mathbf{f}_{\text{disc}}}{\partial \mathbf{x}} \right|_{\mathbf{x}_c, \mathbf{u}_c} \mathbf{E}(\mathbf{q}_c), \quad (19)$$

$$\mathbf{B}_k = \mathbf{E}(\mathbf{q}_c)^\top \left. \frac{\partial \mathbf{f}_{\text{disc}}}{\partial \mathbf{u}} \right|_{\mathbf{x}_c, \mathbf{u}_c}. \quad (20)$$

where

$$\mathbf{E}(\mathbf{q}_c) = \begin{bmatrix} \Xi(\mathbf{q}_c) & & \\ & \mathbf{I}_3 & \\ & & \mathbf{I}_3 \end{bmatrix} \quad (21)$$

handles the reduction from the quaternion space to the tangent error space. The system matrices $\mathbf{A}_k$ and $\mathbf{B}_k$ are time-varying due to the evolving reference quaternion $\mathbf{q}_c$ and the dependence of the magnetorquer effectiveness on the local magnetic field $\mathbf{b}_B$. However, since these parameters evolve slowly relative to the attitude control bandwidth, we employ a Quasi-Static design approach [16]. By treating the time index as a parameter, we solve a frozen-time DARE at each step. To ensure computational efficiency, we update the controller online using a single Newton-Kleinman iteration [17]. Initialized with the solution $\mathbf{P}_{k-1}$ from the previous step, this update efficiently tracks the optimal solution by exploiting the continuous dependence of $\mathbf{P}_k$ on the slowly varying system parameters.

E. MPC

While the Adaptive LQR provides an optimal policy for stabilization and momentum management near the equilibrium, it inherently assumes a linear system model and unconstrained actuation. However, the satellite attitude control problem is governed by nonlinear rotational dynamics and strict actuator saturation limits. To address these concerns we implement a nonlinear Model Predictive Control (MPC) policy.

1) *Formulation:* The control problem is formulated as a finite-horizon Constrained Trajectory Optimization Problem. At each time step, the controller solves for the optimal sequence of control inputs $\mathbf{u}$ that minimizes a cost function subject to the system dynamics and constraints over a horizon N

$$\min_{\mathbf{u}_0, \dots, \mathbf{u}_{N-1}} \ell_N(\mathbf{x}_N) + \sum_{k=0}^{N-1} \ell_k(\mathbf{x}_k, \mathbf{u}_k) \quad (22)$$

$$\text{s.t. } \mathbf{x}_{k+1} = \mathbf{f}_{\text{disc}}(\mathbf{x}_k, \mathbf{u}_k), \quad (23)$$

$$\mathbf{g}(\mathbf{x}_k, \mathbf{u}_k) \leq 0, \quad k = 0, \dots, N-1 \quad (24)$$

and applies the first input $\mathbf{u}_0$ to the system. We choose a quadratic cost function like for the LQR for the stage cost. The terminal cost is made up of just the the state term, i.e.

$$\ell_N(\mathbf{x}_N) = \delta \mathbf{x}_N^T \mathbf{Q}_N \delta \mathbf{x}_N.$$

To solve the optimization problem, we utilize the Iterative LQR (iLQR) algorithm, combined with an Augmented Lagrangian method to handle inequality constraints (AL-iLQR) [18]. This approach lends itself to the adaptations needed for optimizing over quaternions better than using general nonlinear programming (NLP) solvers which inherently rely on the state space being a vector space.

2) *Constraints*: A critical advantage of the MPC formulation is the explicit handling of actuator constraints. We transform the physical limits of the actuators, which are defined in terms of motor currents and wheel speeds, into constraints on the state $\mathbf{x}$ and control input $\mathbf{u}$. Let $\mathbf{\Lambda}_w = [\hat{\mathbf{a}}_1, \hat{\mathbf{a}}_2, \hat{\mathbf{a}}_3]$ be the matrix of reaction wheel spin axes and $\mathbf{D}_w = \text{diag}(J_{w,1}, J_{w,2}, J_{w,3})$ the matrix of wheel inertia. The total wheel angular momentum from (18) is

$$\mathbf{h}_w = \mathbf{\Lambda}_w \mathbf{D}_w (\boldsymbol{\Omega} + \mathbf{\Lambda}_w^T \boldsymbol{\omega}). \quad (25)$$

Solving for the wheel speeds $\boldsymbol{\Omega} = [\Omega_1, \Omega_2, \Omega_3]^T$ yields the constraint inequality:

$$|\boldsymbol{\Omega}| = \left| \mathbf{D}_w^{-1} \mathbf{\Lambda}_w^{-1} \mathbf{h}_w - \mathbf{\Lambda}_w^T \boldsymbol{\omega} \right| \leq \boldsymbol{\Omega}_{\text{sat}}. \quad (26)$$

The torque actuators are current-limited by their driver electronics. By defining $\mathbf{K}_w := \text{diag}(K_{w,1}, K_{w,2}, K_{w,3})$ and inverting the torque relation (7) we get

$$|\mathbf{i}_w| = \left| \mathbf{K}_w^{-1} \mathbf{\Lambda}_w^{-1} \mathbf{u}_w \right| \leq \mathbf{i}_{w,\text{sat}}. \quad (27)$$

For deriving current constraints for the magnetorquers the procedure is similar. We invert the magnetic dipole mapping (15) and get

$$\left| \mathbf{K}_m^{-1} \mathbf{\Lambda}_m^{-1} \left(\mathbf{b}_B \times \frac{\mathbf{u}_m}{\|\mathbf{b}_B\|^2} \right) \right| \leq \mathbf{i}_{m,\text{sat}}.$$

VI. EXPERIMENTS AND RESULTS

This section evaluates the proposed control architectures through three representative simulation experiments using the parameters detailed in Table I. We benchmark the performance of the adaptive methods, specifically the Adaptive LQR and MPC, against a baseline PD controller tuned using the analytic method proposed by Avanzini et al. [4].

A. Implementation and Controller Tuning

To validate our quaternion-based AL-iLQR implementation, we cross-referenced its performance against a "naive" MPC formulation. This benchmark treated the quaternion as a vector in $\mathbb{R}^4$ and was implemented using the CasADi Opti stack with IPOPT [19] as the nonlinear programming (NLP) solver, transforming the optimal control problem via multiple shooting. The validation confirmed that our manifold-respecting optimization yields consistent results while ensuring strictly valid quaternion normalization. While not outperforming the naive approach.

Distinct tuning strategies were employed to highlight the specific strengths of each controller. The weighting matrices for the LQR were selected as

$$\begin{aligned} \mathbf{Q}_{\text{LQR}} &= \text{diag}(5, 5, 5, 2, 2, 2, 700, 700, 700) \\ \mathbf{R}_{\text{LQR}} &= \text{diag}(70, 70, 70, 7, 7, 7) \cdot 10^7, \end{aligned} \quad (28)$$

prioritizing efficient, long-horizon regulation. In contrast, the MPC was tuned for more aggressive performance closer to the actuator saturation limits, using

$$\begin{aligned} \mathbf{Q}_{\text{MPC}} &= \text{diag}(8, 8, 8, 2, 2, 2, 20, 20, 20) \\ \mathbf{R}_{\text{MPC}} &= \text{diag}(30, 30, 30, 3, 3, 3) \cdot 10^4, \end{aligned} \quad (29)$$

with a terminal cost of $\mathbf{Q}_N = 20 \cdot \mathbf{Q}_{\text{MPC}}$. We observed that the MPC does not perform well in low-energy regimes, whereas the LQR shows good performance. In these scenarios, the high penalty on actuation results in slow system dynamics with time constants that far exceed the MPC's standard prediction horizon. The infinite-horizon nature of the LQR naturally captures the full evolution of these slow trajectories, giving it an advantage over the finite-horizon MPC, which suffers from the horizon truncation without a perfect terminal cost approximation.

B. Performance Metrics

Our quantitative assessment relies on three primary metrics. First, the *actuator effort* e quantifies the energy expenditure and is defined as

$$e = \sum_i \int i_i^2 dt, \quad (30)$$

where i represents the actuator index. It is important to note that while e is proportional to energy, it does not permit a direct comparison between magnetorquers and reaction wheels due to their differing electrical resistances. Second, we examine the *attitude error* between the orbital frame $\mathcal{O}$ and the body frame $\mathcal{B}$ utilizing 2-1-3 Euler angles. Finally, the *settling time* denotes the duration required to reach and maintain an attitude error within a $\pm 1^\circ$ band.

C. Nominal Performance Analysis

We first assess the system under nominal conditions with perfect state estimation and no parametric uncertainty. Experiment 1 focuses on a standard detumbling maneuver, where the spacecraft is initialized with attitude errors of $[10^\circ, -10^\circ, 10^\circ]$ and angular rates of $\boldsymbol{\omega} = [1, 1, -1]^T$ °/s.

The adaptive control architectures achieved an approximate 15% reduction in actuator effort of both magnetorquers and reaction wheels compared to the baseline PD controller, while maintaining a comparable or even slightly reduced settling time. This efficiency gain is attributed to the adaptive controllers' capacity to exploit real-time knowledge of the magnetic field vector and their utilization of additional degrees of freedom within the feedback law. The corresponding attitude evolution is depicted in Fig. 2. While the MPC achieves significantly faster settling times than the other architectures, it does so at the cost of higher actuation effort. Furthermore, close inspection of Fig. 2c reveals a slight steady-state pitch angle offset. This deviation is likely

caused by unmodeled environmental disturbances, such as solar radiation pressure (SRP) and aerodynamic drag. This effect becomes more pronounced when actuation effort is penalized less heavily in the cost function. Although the MPC's aggressive tuning partially mitigates these errors, complete elimination of the offset would require integrating the disturbance torques directly into the controller's internal model.

The second experiment investigated large-angle stabilization. The initial attitude errors were $[-100^\circ, 80^\circ, 100^\circ]$ with initial angular velocities of $\omega = [1, 1, -1]^\top$ °/s. While all controllers maintained stability, the performance advantage of the Adaptive LQR over the baseline was not observed in this regime. In fact, the LQR actuator effort increased to approximately double that of the baseline. This performance degradation occurs because the LQR optimality is predicated on the validity of the linearization around the equilibrium trajectory, a condition that is violated during large-angle maneuvers where the majority of the control effort is expended.

D. Robustness to Uncertainty

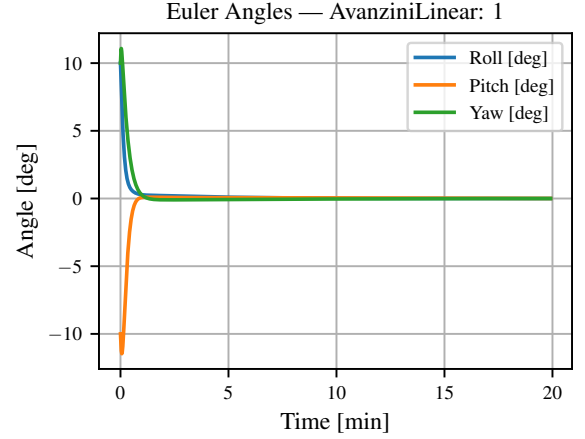
The final experimental phase evaluates robustness against stochastic effects. To assess performance under uncertainty, we conducted a series of randomized simulations where all controllers were tested against identical seeds, with metrics averaged over 5 runs. Parametric uncertainty was introduced as a uniform distribution with $\pm 10\%$ bounds applied to both the reaction wheel spin inertia and the diagonal elements of the spacecraft inertia tensor.

Figure 3 illustrates the attitude evolution for a representative simulation run. All controllers remained stable and exhibited behavior similar to the nominal case. However, a distinct trade-off in control allocation was observed. The adaptive controllers (LQR and MPC) utilized the magnetorquers more aggressively, resulting in effort metrics twice as large as the baseline, while significantly reducing the load on the reaction wheels (approximately half the effort of the baseline). This suggests that the adaptive schemes are more effective at exploiting the magnetorquers for stabilizing the satellite as opposed to the baseline controller which relies solely on the reaction wheels.

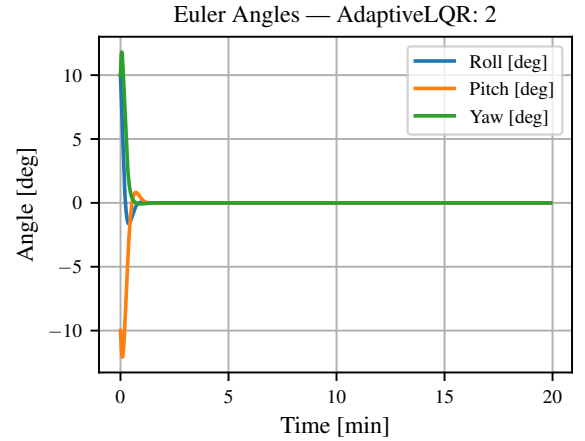
VII. CONCLUSIONS AND OUTLOOK

This study demonstrated that unified control architectures can significantly improve actuation efficiency for small satellite by coordinating attitude stabilization with momentum management. The proposed Adaptive LQR and nonlinear MPC approaches successfully exploited magnetic torques to more than just momentum dumping. Notably, the LQR architecture reduced effort by approximately 15% in the nominal case, while the MPC demonstrated very fast convergence. However, simulation results also highlighted limitations in their current formulations that necessitate further development.

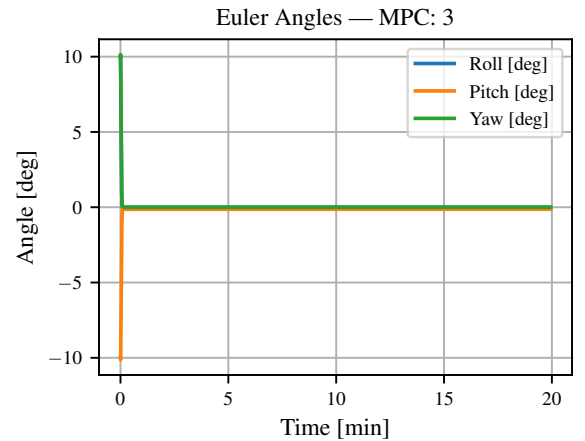
The persistent steady-state offset observed in the MPC highlights its sensitivity to unmodeled environmental forces. Future implementations must augment the system model to include disturbance torques. By estimating these online via



(a) Baseline Linear Controller

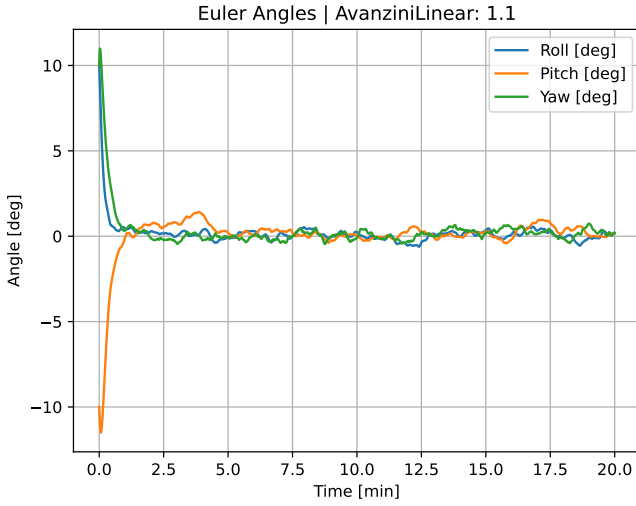


(b) Adaptive LQR Controller

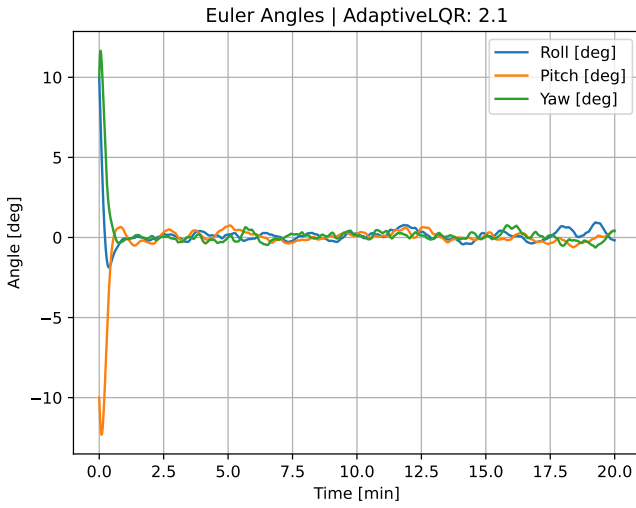


(c) Model Predictive Controller

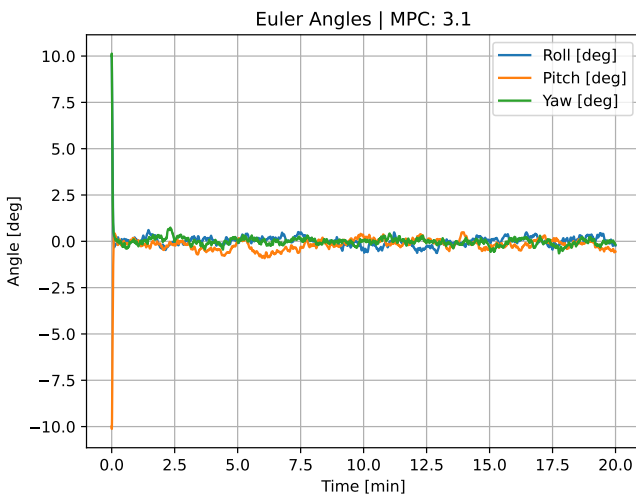
Fig. 2: Evolution of the $\mathcal{O}$ to $\mathcal{B}$ attitude error (2-1-3 Euler angles) for the nominal detumbling scenario with perfect estimation.



(a) Baseline Linear Controller



(b) Adaptive LQR Controller



(c) Model Predictive Controller

Fig. 3: Evolution of the $\mathcal{O}$ to $\mathcal{B}$ attitude error (2-1-3 Euler angles) for the detumbling scenario with real estimation and parametric uncertainty.

a disturbance observer, the MPC could compensate for them directly, eliminating steady-state errors without relying on integral action.

Furthermore, while the "frozen-time" assumption proved valid for nadir pointing, it may become untenable for fast-varying reference trajectories. Future research should evaluate the breakdown point of the Adaptive LQR against the nonlinear MPC approach, which should better suited to handle such aggressive maneuvers.

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Parameter	Symbol	Value	Unit
Mass	m	2.304	kg
Inertia	$\mathbf{J}_B$	$\begin{bmatrix} 12.286 & -0.065 & -0.236 \\ -0.065 & 12.726 & -0.039 \\ -0.236 & -0.039 & 5.707 \end{bmatrix}$	g m^2
Sim. time step	Δt	0.2	s
Starting datetime	t_0	2024-01-01, 12:00:00	UTC
Inclination	i	97.6	deg
RAAN	Ω	10.0	deg
Eccentricity	e	0.0001	–
Arg. of Perigee	ω	0.0	deg
Mean Anomaly	M	0.0	deg
Mean Motion	n	15.25	rev/day
Axial inertia	$J_{w,i}$	2.82×10^{-6}	kg m^2
Max torque	$\tau_{\max,i}$	2.0×10^{-3}	N m
Max speed	$\Omega_{\max,i}$	628.319	rad/s
Max current	$i_{\max,i}$	1.0	A
Current time cons.	$T_{\text{cur},i}$	0.1	s
Torque cons.	$K_{w,i}$	2.0×10^{-3}	N m/A
Max dipole mom.	$m_{\max,j}$	(0.3, 0.3, 0.2)	A m^2
Max current	$i_{\max,m,j}$	0.3	A
Current time cons.	$T_{\text{cur},m,j}$	0.2	s
Moment cons.	$K_{m,j}$	(1, 1, 0.667)	m^2

TABLE I: Parameters of the simulation which stayed constant for all experiments.

APPENDIX

a) *Actuator parameters.*: Table I summarizes the actuator limits and first-order current dynamics used in all simulations (see simulation configuration).

The torque and dipole constants are derived from the saturation limits, $K_{w,i} = \tau_{\max,i}/i_{\max,i}$ and $K_{m,j} = m_{\max,j}/i_{\max,m,j}$.

A. Mathematical Framework

This section gives a more in depth look at the algebraic properties of unit quaternions and the differential calculus techniques required for optimization on the quaternion manifold. The information is mostly gathered from [2] for the quaternion operators and [3] for the differential calculus.

1) *Unit Quaternions*: A unit quaternion $\mathbf{q} \in \mathbb{H}$ is defined by a vector part $\mathbf{q}_{1:3} \in \mathbb{R}^3$ and a scalar part $q_4 \in \mathbb{R}$, denoted as:

$$\mathbf{q} = \begin{bmatrix} \mathbf{q}_{1:3} \\ q_4 \end{bmatrix},$$

satisfying the unity constraint $\|\mathbf{q}\|_2 = 1$. Quaternion multiplication¹ is defined as:

$$\bar{\mathbf{q}} \otimes \mathbf{q} = \begin{bmatrix} q_4 \bar{\mathbf{q}}_{1:3} + \bar{q}_4 \mathbf{q}_{1:3} - \bar{\mathbf{q}}_{1:3} \times \mathbf{q}_{1:3} \\ \bar{q}_4 q_4 - \bar{\mathbf{q}}_{1:3} \cdot \mathbf{q}_{1:3} \end{bmatrix}.$$

Rotations of a vector $\mathbf{v}$ from frame $\mathcal{F}$ to frame $\mathcal{G}$ are computed via conjugation:

$$\begin{bmatrix} \mathbf{v}_G \\ 0 \end{bmatrix} = \mathbf{q}_{GF} \otimes \begin{bmatrix} \mathbf{v}_F \\ 0 \end{bmatrix} \otimes \mathbf{q}_{GF}^*,$$

where $\mathbf{q}^* = [-\mathbf{q}_{1:3}^\top, q_4]^\top$ denotes the conjugate quaternion.

¹We adopt the JPL convention as standard in aerospace applications [14]. However, very uncommon in most other applications and libraries where the Hamilton convention is preferred.

2) *Quaternion Differential Calculus*: Standard gradient-based optimization is ill-suited for quaternions due to the unit-norm constraint. The derivatives lie in the tangent space of the unit hypersphere. We denote differential rotations in this tangent space as vectors $\phi \in \mathbb{R}^3$.

While several mappings exist, we parameterize local perturbations using the vector part of the quaternion. The mapping $\varphi: \mathbb{R}^3 \rightarrow \mathbb{H}$ and its inverse are given by:

$$\mathbf{q} = \varphi(\phi) = \begin{bmatrix} \phi \\ \sqrt{1 - \phi^\top \phi} \end{bmatrix}, \quad \phi = \varphi^{-1}(\mathbf{q}) = \text{sgn}(q_4) \mathbf{q}_{1:3}.$$

An infinitesimal rotational perturbation is approximated as $\delta \mathbf{q} \approx [\phi^\top, 1]^\top$. Applying this perturbation to a nominal attitude $\mathbf{q}$ yields

$$\delta \mathbf{q} \otimes \mathbf{q} \approx \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} \phi \\ 0 \end{bmatrix} \right) \otimes \mathbf{q} = \mathbf{q} + \Xi(\mathbf{q})\phi.$$

The matrix $\Xi(\mathbf{q})$, referred to as the attitude Jacobian, is the linearization of the map from tangent space $\mathbb{R}^3$ to quaternion space $\mathbb{H}$

$$\Xi(\mathbf{q}) \triangleq \begin{bmatrix} q_4 \mathbf{I}_3 + [\mathbf{q}_{1:3} \times] \\ -\mathbf{q}_{1:3} \end{bmatrix}. \quad (31)$$

This matrix allows us to define the Jacobian of a vector-valued function $\mathbf{h}(\mathbf{q}): \mathbb{H} \rightarrow \mathbb{R}^p$ with respect to the local perturbation ϕ :

$$\nabla \mathbf{h}(\mathbf{q}) = \frac{\partial \mathbf{h}}{\partial \mathbf{q}} \Xi(\mathbf{q}) \in \mathbb{R}^{p \times 3}. \quad (32)$$

For a scalar (cost) function ($p = 1$), the Hessian is computed as:

$$\nabla^2 \mathbf{h}(\mathbf{q}) = \Xi(\mathbf{q})^\top \frac{\partial^2 \mathbf{h}}{\partial \mathbf{q}^2} \Xi(\mathbf{q}) - \mathbf{I}_3 \left(\frac{\partial \mathbf{h}}{\partial \mathbf{q}} \mathbf{q} \right) \in \mathbb{R}^{3 \times 3}.$$

Finally, for quaternion-valued functions $\mathbf{f}(\mathbf{q}): \mathbb{H} \rightarrow \mathbb{H}$ (e.g., discrete dynamics), the Jacobian projects the input perturbation to the output tangent space:

$$\nabla \mathbf{f}(\mathbf{q}) = \Xi(\mathbf{f}(\mathbf{q}))^\top \frac{\partial \mathbf{f}}{\partial \mathbf{q}} \Xi(\mathbf{q}) \in \mathbb{R}^{3 \times 3}. \quad (33)$$

B. Extended Kalman Filter (EKF) Details

The EKF estimates spacecraft attitude and gyroscope bias. The filter is implemented in an error-state form with

$$\delta \mathbf{x} = \begin{bmatrix} \delta \boldsymbol{\theta} \\ \delta \mathbf{b}_\omega \end{bmatrix} \in \mathbb{R}^6, \quad \mathbf{P} = \mathbb{E}[\delta \mathbf{x} \delta \mathbf{x}^\top], \quad (34)$$

where $\delta \boldsymbol{\theta}$ is a small-angle attitude error and $\delta \mathbf{b}_\omega$ is the gyro bias error.

1) *Prediction*: The bias-compensated angular rate is

$$\tilde{\boldsymbol{\omega}} = \boldsymbol{\omega}_{\text{mea}} - \tilde{\mathbf{b}}_\omega, \quad (35)$$

and the quaternion is propagated using the kinematic equation. The linearized continuous-time error dynamics are

$$\dot{\delta \mathbf{x}} = \mathbf{F} \delta \mathbf{x} + \mathbf{G} \mathbf{w}, \quad (36)$$

with

$$\mathbf{F} = \begin{bmatrix} -[\tilde{\boldsymbol{\omega}}]_\times & -\mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}. \quad (37)$$

A first-order discretization is used:

$$\Phi \approx \mathbf{I}_6 + \mathbf{F} \Delta t, \quad \mathbf{P}_{k+1}^- = \Phi \mathbf{P}_k \Phi^\top + \mathbf{Q}_d. \quad (38)$$

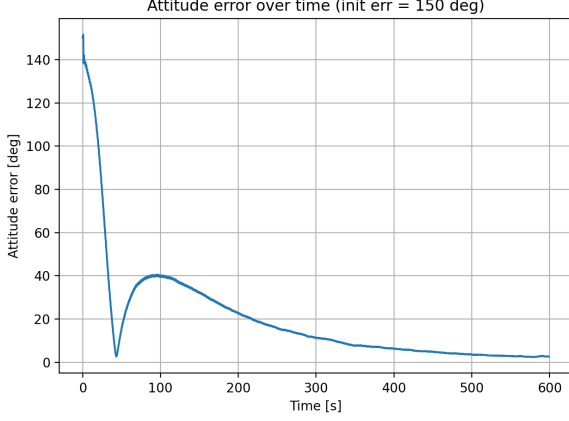


Fig. 4: EKF attitude error over time for the worst-case initial offset (150°).

Parameter	Symbol	Value
Gyro noise std. dev.	σ_ω	1.0×10^{-3} rad/s
Gyro bias rand. walk std. dev.	σ_b	3.16×10^{-6} rad/s/ $\sqrt{s}$
Sun s. unit-vec. noise std. dev.	σ_{sun}	1.0×10^{-2} ($\approx 0.57^\circ$)
Mag. unit-vec. noise std. dev.	σ_{mag}	2.0×10^{-3} ($\approx 0.115^\circ$)
Initial attitude covariance	$\mathbf{P}_{0,\theta}$	$(5 \times 10^{-2})^2 \mathbf{I}_3$
Initial bias covariance	$\mathbf{P}_{0,b}$	$(1 \times 10^{-3})^2 \mathbf{I}_3$

TABLE II: EKF parameters used in simulation. Sensor variances are taken from `simulation_config.json`. The magnetometer unit-vector noise is approximated assuming a nominal $\|\mathbf{b}\| \approx 50 \mu\text{T}$ in LEO.

2) *Vector measurement updates (Sun sensor, magnetometer)*: Both sensors provide direction measurements in the body frame. Given an inertial reference vector $\mathbf{z}_I$, the predicted body-frame direction is

$$\tilde{\mathbf{z}}_B = \mathbf{R}(\tilde{\mathbf{q}}) \mathbf{z}_I, \quad \mathbf{y} = \mathbf{z}_{B,\text{mea}} - \tilde{\mathbf{z}}_B. \quad (39)$$

The corresponding Jacobian used in the implementation is

$$\mathbf{H} = \begin{bmatrix} [\tilde{\mathbf{z}}_B]_\times & \mathbf{0}_3 \end{bmatrix}. \quad (40)$$

Note that the sign of the cross-product term depends on the chosen multiplicative attitude-error convention; the implementation follows the right-multiplicative (JPL) convention. During eclipse, the Sun sensor measurement is unavailable; the Sun update is skipped and the EKF continues with gyro propagation and magnetometer updates.

Fig. 4 shows the EKF convergence in the worst-case initialization scenario.

3) *Filter tuning and initialization*: The EKF performance depends on the assumed sensor noise and bias dynamics. We use a diagonal process noise model with gyroscope white noise and gyro-bias random walk. All covariance values below match the implementation used in simulation.

In the correction step, the estimated attitude is updated multiplicatively using the small-angle error $\delta\boldsymbol{\theta}$ via $\delta\mathbf{q} \approx [\frac{1}{2}\delta\boldsymbol{\theta}^\top, 1]^\top$ (JPL convention), followed by quaternion renormalization. All vector measurements are normalized to unit length before forming the residual.